

Harvest Hold Rwanda Prototype Materials & Estimated Budget

Harvest Hold Rwanda – Smart Hybrid Charcoal Cooling System (SHCCS)

An IoT and AI-powered agricultural cooling prototype designed to reduce post-harvest losses in Rwanda through affordable and sustainable cooling technology.

Prototype Materials and Estimated Costs

No	Material / Component	Quantity	Estimated Unit Price (RWF)	Total Price (RWF)	Purpose
1	ESP32 DevKit V4	1	14,000	14,000	Main microcontroller for IoT system
2	SHT3x Temperature & Humidity Sensor	1	14,500	14,500	Environmental monitoring
3	Water Drip Kit / Small Pump	1	10,000	10,000	Wet charcoal cooling
4	Small Solar Panel 12V	1	35,000	35,000	Off-grid power supply
5	Small rechargeable battery	1	15,000	15,000	Power backup
6	Plastic/Wooden Prototype Box	1	25,000	25,000	Prototype structure
7	DC Cooling Fan	1	8,000	8,000	Air circulation
8	PVC Pipe & Small Water Container	1 set	8,000	8,000	Water distribution
10	Charcoal Cooling surface	1 set	15,000	15,000	Evaporative cooling
14	Relay Module	1	7,000	7,000	Fan and pump automation

15	Jumper Wires & Connectors	1 set	3,000	3,000	Electronic Connections
16	Breadboards / PCB (I think option is the best option)	2	2,000	4,000	Circuit prototyping

Estimated Total Budget

158,500 RWF

Additional Notes

- ☐ Prices are estimated based on available electronic and hardware markets in Rwanda and online suppliers.
- ☐ Some components may vary in price depending on importation and quality.
- ☐ Locally sourced materials such as charcoal, wood, and water systems help reduce implementation costs.
- ☐ The prototype is designed to operate fully off-grid using solar energy and low-water evaporative cooling systems.
- ☐ The system integrates IoT sensors and cloud-based monitoring for smart agricultural storage management.